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Global Managerial Competence, Global Entrepreneurial Orientation, and Sustainable International Expansion Outcomes: Evidence from Emerging Market Born Global Firms

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Abstract: Born global firms (BGFs) play a significant role in the expansion of sustainable industries through their rapid internationalisation and early adoption of innovation-oriented organisational frameworks. However, in emerging markets, sustainable global expansion is often constrained by organisational inefficiencies, infrastructural challenges, and intense international competition. In this context, the present study examines the influence of Global Entrepreneurial Orientation (GEO) and Global Managerial Competence (GMC) on Sustainable International Expansion Outcomes (SIEO), aligned with the objectives of Sustainable Development Goal (SDG) 9 (Industry, Innovation, and Infrastructure). The study is grounded in the Resource-Based View (RBV) and Dynamic Capabilities Theory (DCT), which explain how firms develop, integrate, and utilise strategic capabilities to achieve competitiveness in international markets. The proposed conceptual framework is empirically tested using Partial Least Squares Structural Equation Modelling (PLS-SEM) based on primary data collected from 430 executives and administrators of international firms operating in India, Singapore, Malaysia, and the United Arab Emirates. The findings reveal that GEO significantly enhances GMC and positively influences SIEO. Furthermore, GMC significantly moderates the relationship between GEO and SIEO. The negative interaction effect indicates a buffering pattern, whereby the positive influence of GEO on SIEO becomes less pronounced at higher levels of managerial competence. These findings demonstrate that entrepreneurial orientation and managerial competence interact in shaping the sustainable international expansion of BGFs. This study contributes to the international entrepreneurship literature by integrating sustainability perspectives with strategic orientation, managerial capability, and international performance. The findings also provide valuable managerial and policy implications for business leaders and policymakers in emerging economies seeking to foster sustainable international growth.

Keywords: Global Managerial Competence; Sustainable International Expansion Outcomes; Sustainable Development Goal 9; Emerging economies; Global Entrepreneurial Orientation

1. Introduction

The rapid international expansion of new firms has reshaped patterns of business growth and intensified global competition. Born global firms (BGFs) enter international markets within a short period after their establishment and generate a substantial share of their revenue from foreign operations, distinguishing them from traditional step-by-step internationalization models (Knight & Cavusgil, 2004; Oviatt & McDougall, 2004). Advances in technology, improvements in global logistics, and the expansion of information networks have enabled small firms with limited resources to participate and compete in international markets to an extent never seen before (Gabrielsson & Kirpalani, 2004; Rialp et al., 2005). Born-global firms now drive innovation and sustainable growth in industry worldwide. Research in transnational business and entrepreneurship has increasingly focused on sustainability, with particular attention to Sustainable Development Goal (SDG)-9, which promotes inclusive industrial development, resilient infrastructure, and innovation-driven production (Loorbach et al., 2017; United Nations, 2015). Emerging economies struggle with infrastructure, legal issues, and limited international research networks (Peng & Luo, 2000; Prashantham & Young, 2011). Born-global firms combine regional creativity with worldwide supply chains for ecologically sustainable industrial improvement and technological dissemination (Autio, 2017; Bocken et al., 2014). An early and proactive globalisation strategy includes a firm's worldwide Entrepreneurial Approach. GEO reflects how a firm applies creativity, strategic thinking, and a willingness to take risks to identify and make the most of opportunities in international markets (Knight, 2000; Lumpkin & Dess, 1996). International entrepreneurs with a strong focus on innovation are more inclined to introduce new products, enter foreign markets, and compete actively with established firms (Boso et al., 2012; Jantunen et al., 2008). GEO supports SDG-9 of sustainability by promoting innovation, technological upgrades, and resource optimisation across borders (Autio, 2017; Lee & Trimi, 2018). Entrepreneurial orientation alone does not ensure successful international expansion. The ability of a firm to translate entrepreneurial intent into sustained international outcomes also depends on the competence of its managers (Martin & Javalgi, 2016; Wiklund & Shepherd, 2005). BGFs require managers with international business knowledge, cross-cultural competence, global networking capabilities, strategic decision-making skills, and the ability to adapt organisational resources to changing market conditions. Such managerial competencies are particularly important in emerging economies, where firms frequently operate under conditions of institutional uncertainty, infrastructural constraints, and limited access to international knowledge networks.

Although research on BGFs and sustainable internationalisation has grown, limited empirical attention has been devoted to how managerial competence conditions the effectiveness of entrepreneurial orientation in generating Sustainable International Expansion Outcomes (SIEO), particularly in emerging economies. Existing studies have largely examined entrepreneurial orientation and managerial capabilities as direct determinants of international performance, leaving the interaction between these strategic and managerial capabilities insufficiently understood. Drawing on the Resource-Based View (RBV) and Dynamic Capabilities Theory (DCT), this study therefore examines whether Global Managerial Competence (GMC) moderates the relationship between Global Entrepreneurial Orientation (GEO) and SIEO. By focusing on BGFs in India, Malaysia, Singapore, and the United Arab Emirates, the study provides a capability-based explanation of how the effectiveness of entrepreneurial orientation may vary across different levels of managerial competence. As per de Sousa et al. (2024) and Sinkovics et al. (2016), managerial abilities and innovative thinking are autonomous factors and not included in a sustainability-based competency framework applicable to SDG-9. How administrative global competency mediates the connection between an entrepreneurial attitude and sustainable overseas outcomes in developing nations is unclear. This study creates, empirically tests, and evaluates a comprehensive conceptual framework that connects Global Business Orientation, managerial Global proficiency, and Responsible International Expansion Results within resource-based approaches and DCT to close gaps. This study examines primary data collected from BGFs operating in India, Malaysia, Singapore, and the United Arab Emirates to assess how GEO influences sustainability-oriented international performance, both in the short and long term, through the role of GMC. This study adopts SDG 9 as a sustainability reference framework to advance international entrepreneurship theory and to generate policy-relevant insights for fostering innovation-led development and sustainable global expansion in emerging markets.

This study offers four key contributions to the literature on sustainability and entrepreneurship. First, it advances understanding of sustainable foreign development outcomes by aligning global entrepreneurial practices with the objectives of SDG 9, thereby integrating sustainability considerations into entrepreneurial orientation and international performance. Second, it demonstrates that GMC plays a mediating role between entrepreneurial strategy and sustained international outcomes, clarifying the mechanism through which strategic entrepreneurship translates into global performance. The study is supported by empirical evidence from 430 firms classified as born global, operating across four emerging economies, namely India, Malaysia, Singapore, and the United Arab Emirates. By focusing on these contexts, the research addresses a notable gap in global entrepreneurship literature, which has predominantly emphasized developed economies. The PLS-SEM mediation paradigm supports sustainability-driven internationalisation beyond direct effects. The study extends the capability-based understanding of sustainable international expansion by examining GMC as a boundary condition in the GEO–SIEO relationship. Rather than treating managerial competence as an intervening mechanism, the study demonstrates that the contribution of entrepreneurial orientation to SIEO varies according to the level of GMC. This moderation perspective provides a more nuanced explanation of how strategic entrepreneurial orientation and managerial capabilities jointly shape international expansion outcomes. Five sections comprise this study. After reviewing the existing literature, the research hypotheses are formulated. The study subsequently details the analytical approach, reports the core findings, and closes by outlining policy recommendations and directions for follow-up research.

2. Literature Review

The research on international entrepreneurship acknowledges that globalisation requires innovation-driven performance, management skills, and strategic orientation. Born-global firms often expand internationally at an early stage despite operating in uncertain environments and facing resource limitations. Their ability to sustain long-term international growth largely depends on the development of intangible capabilities such as managerial competence and entrepreneurial orientation (Knight & Cavusgil, 2004; Oviatt & McDougall, 2004). The RBV and DCT provide a useful framework for understanding how these internal competencies support sustained global expansion (Barney, 1991; Teece et al., 1997). SDG-9 requires ongoing industrialisation, innovation, and strong infrastructure for global economic growth (United Nations, 2015). International entrepreneurship supports knowledge transmission, technical transformation, and technology shift, becoming critical to this goal (Autio, 2017; Loorbach et al., 2017). Eisenhardt & Martin (2017) and Sinkovics et al. (2016) state that sustainable worldwide success requires market entrance speed and the ability to reorganise resources and talents in changing global settings. Global energetic orientation, or GEO, has been extensively studied because it encourages creativity, readiness, and threat engaging in the international marketplace (Knight, 2000; Lumpkin & Dess, 1996). Managerial global capability has been conceptualized as a dynamic organizational capability that enables firms to translate entrepreneurial intentions into successful international performance (Man et al., 2002; Ang et al., 2007). In the context of SDG-9, SIEO encompass dimensions such as financial sustainability, innovation continuity, organizational resilience, and efficiency in infrastructure and operational development (Bocken et al., 2014).

2.1 Conceptual and Theoretical Framework

The RBV and DCT explain how organisations obtain and maintain their competitive edge in dynamic and uncertain environments. According to the RBV, precious, uncommon, unique, and indeterminate (VRIN) resources provide companies with an advantage over competitors (Barney, 1991; Wernerfelt, 1984). As inborn-global firms lack tangible resources, entrepreneurship and managerial skills are vital for their worldwide competitiveness (Autio et al., 2000; Peng & Luo, 2000). The RBV alone cannot adequately explain competitive advantage in dynamic global environments. DCT highlights how firms adapt by integrating and reconfiguring resources in response to change. For BGFs, especially in volatile international markets, dynamic competencies such as global management are essential for identifying and exploiting opportunities. A global entrepreneurial mindset thus acts as a key strategic capability influencing international opportunity recognition and creation (Jantunen et al., 2008; Knight, 2000). GEO firms conduct proactive market analysis, seek foreign potential, and allocate funds to uncertain global initiatives. GEO increases strategic renewal detection and implementation for sustainable internationalisation (Autio, 2017; Teece, 2014). Implementing entrepreneurial approaches in diverse organisational and cultural situations requires dynamic managerial international competence (Ang et al., 2007; Man et al., 2002). Increasing managers' cultural knowledge, global networking, and flexible thinking helps organisations overcome uncertainty and newness (Coviello & Munro, 1997; Zaheer, 1995). SDG-9-aligned innovation-driven projects require these competencies for sustainable industry and infrastructure growth (Bocken et al., 2014). SIEO demonstrates how these essential resources and abilities will impact success. Innovation quantity, global learning, building efficiency, and international company resilience are SIEO's alternative financial success metrics (Kuivalainen et al., 2012; Sinkovics et al., 2016). SDG-9 performance review describes sustainable global success as profit-generating, technological advancement, and equitable growth (de Sousa et al., 2024; United Nations, 2015). Drawing on the RBV and DCT, this study conceptualises GEO as a strategic behavioural capability, GMC as an adaptive managerial capability, and SIEO as the sustainability-oriented outcomes of international growth. The framework proposes that entrepreneurial orientation directly contributes to sustainable international expansion, while managerial competence both directly influences SIEO and conditions the strength of the GEO–SIEO relationship. This perspective provides a capability-based explanation of how entrepreneurial and managerial resources jointly shape sustainable international expansion among emerging-market- BGFs. Emerging-market born-global firms need adaptive skill development due to institutional ambiguities and infrastructural constraints (Prashantham & Young, 2011; Teece, 2014).

2.2 Dynamic Managerial Capabilities: The Cognitive, Social, and Human Aspects of Management

Managers may swiftly construct, integrate, and reconfigure firm resources and competencies using Dynamic Managerial Capabilities (DMC) (Teece, 2014). In global-born enterprises, where a small management team or one company owner makes tactical decisions, managerial abilities disproportionately affect global planning and performance that is sustainable. DCT-based DMC helps organisations discover global opportunities, invest wisely, and transform resources to foster long-term international growth. Knowledge, skills, education, and global experience help managers make strategic decisions. Managerial human capital enables BGFs to comprehend global market dynamics, technological developments, and competitiveness (Autio et al., 2000). Managers with international competence may implement new strategies, coordinate operations across borders, and align growth with SDG-9 (de Sousa et al., 2024). Managers' social capital includes suppliers, customers, investors, and overseas partners. Social capital enables born-global companies to acquire foreign market expertise, innovations, and distribution connections (Coviello & Munro, 1997). Strong international networks encourage green innovation dissemination and global value chain involvement, supporting SDG-9 (Autio, 2017; Bocken et al., 2014). Management cognition comprises assessing international market knowledge, possibilities, hazards, and strategies. Cognitive competence influences entrepreneurs' sustainable development, technical, and competitive views in highly unpredictable global contexts (Teece, 2014). Directors with strong cognitive capabilities are better able to balance immediate financial performance with long-term innovation goals in international markets. DMC, including staffing, social influence, and managerial cognition, enable born-global firms to sustain success in global markets. In developing economies, such adaptive managerial capabilities are particularly important for achieving economic growth aligned with SDG-9, given prevailing institutional and infrastructural constraints (Prashantham & Young, 2011; Teece, 2014).

2.3 Entrepreneurial Orientation: Innovation, Proactive Behaviour, and Risk Taking

Entrepreneurial orientation suggests an organisation's tactical posture toward entrepreneurial activity and is commonly characterized by innovativeness, proactiveness, and risk-taking. When applied in an international context, entrepreneurial orientation evolves into GEO, which represents a firm's willingness to pursue cross-border opportunities through innovation-led and risk-aware strategies. Innovativeness supports creativity, technological advancement, and the development of new products, services, and processes, enabling BGFs to compete effectively with larger multinational enterprises (Autio et al., 2000; Gabrielsson & Gabrielsson, 2011; Lumpkin & Dess, 1996). Such innovation-oriented behaviour also contributes to technological progress and sustainable industrial development aligned with SDG 9 (United Nations, 2015). Proactiveness captures a firm's forward-looking orientation and first-mover behavior in global markets, allowing early identification and exploitation of international opportunities in highly dynamic environments (Boso et al., 2012; Knight & Cavusgil, 2004; Lumpkin & Dess, 1996). For BGFs, proactiveness is therefore essential for achieving sustainable growth amid rapid technological change and environmental uncertainty (Autio, 2017).

A firm's inclination to invest substantially in uncertain overseas operations and to tolerate the possibility of failure in order to enhance returns reflects risk taking (Wiklund & Shepherd, 2005). In emerging economies, international expansion is associated with considerable financial, political, and cultural risks (Peng & Luo, 2000; Zaheer, 1995). Therefore, in keeping with SDG-9, sustained global growth necessitates a balanced approach to risk-taking that fosters innovation while guaranteeing long-term resource efficiency and stability (de Sousa et al., 2024; Sinkovics et al., 2016). GEO has been shown to enhance global performance, export intensity, and responsiveness to foreign markets. However, the effectiveness of GEO depends on the presence of complementary managerial capabilities that support the implementation of entrepreneurial strategies in complex international environments (Martin & Javalgi, 2016; Wiklund & Shepherd, 2005). This highlights the importance of linking GEO with adaptive managerial capabilities to achieve sustainable global expansion.

2.4 Operational Capabilities: Marketing and Managerial Capabilities

Operational Capabilities refer to a firm's capacity to carry out routine day-to-day activities and convert strategic goals into measurable performance outcomes (Winter, 2003). Because BGFs operate with constrained resources while expanding rapidly across international markets, they must be capable of implementing strategies effectively across borders (Autio et al., 2000; Knight & Cavusgil, 2004). In this context, operational capabilities are reflected in marketing and managerial competencies, which have repeatedly been shown to play a central role in enhancing international performance. Marketing capabilities reflect how effectively a firm understands customer needs, evaluates competitors, manages pricing, builds branding strategies, and coordinates international distribution channels. Strong marketing capabilities enable BGFs to enter foreign markets, attract international customers, and build brand legitimacy across borders (Kuivalainen et al., 2012; Zou et al., 1998). Marketing functions play a crucial role in facilitating eco-friendly innovation, promoting environmentally sustainable products, and encouraging responsible consumption behaviors in the global marketplace (Bocken et al., 2014; de Sousa et al., 2024). Management capabilities include internal coordination, financial oversight, human resource administration, strategic forecasting and performance supervision. For BGFs, robust managerial capabilities support efficient allocation of limited resources, effective management of international operations, and resilience in uncertain global environments (Martin & Javalgi, 2016). These capabilities are essential for translating managerial vision and entrepreneurial behavior into sustained international performance aligned with SDG-9 (Sinkovics et al., 2016; Teece, 2014). Within the DCT framework, operational capabilities serve as the primary mechanism through which higher-order capabilities such as GEO and Managerial Global Capabilities influence firm performance. Accordingly, this study considers communication and leadership capabilities as key channels linking entrepreneurial and managerial competencies to sustained international expansion outcomes.

2.5 Competitive Intensity

Competitive intensity refers to the degree of rivalry, aggressiveness, and pressure present in a firm's external environment. International markets, particularly in developing economies, experience heightened competition due to global competitors, rapid technological advancements, and evolving customer preferences (Peng & Luo, 2000; Prashantham & Young, 2011). In intensely competitive global niches, BGFs need to constantly innovate and adjust their strategies to remain viable and sustain their presence (Autio, 2017; Knight & Cavusgil, 2004). Elevated competitive intensity demands strong entrepreneurial orientation, managerial flexibility, and efficient operational execution (Martin & Javalgi, 2016; Wiklund & Shepherd, 2005). To sustain competitive advantage, firms are required to regularly adjust their products, business models, and international activities (Eisenhardt & Martin, 2017; Teece, 2014). Under these conditions, GEO and Managerial Global Capabilities become critical for managing uncertainty and supporting long-term growth. Moreover, competitive pressure shapes how organisations pursue innovation and technological development in alignment with SDG-9 from a sustainability perspective. Intense market competition often pushes firms to adopt cleaner technologies, strengthen digital infrastructure, and improve processes to reduce costs while meeting environmental responsibilities (Bocken et al., 2014). However, when firms lack adequate managerial and operational capabilities, intense competitive pressure may strain organisational resources and threaten long-term sustainability (de Sousa et al., 2024; Sinkovics et al., 2016). Therefore, competitive intensity is widely viewed as an important contextual condition that can either strengthen or weaken the influence of business orientation, managerial capabilities, and operational efficiency on sustained international performance. In line with this perspective, the present study incorporates competitive intensity to capture the dynamic external environment within which born-global firms pursue sustainable international expansion. While competitive intensity and operational capabilities are frequently identified as important determinants of international business performance, the present study does not examine these constructs directly. Instead, they are acknowledged as contextual conditions that may influence the broader internationalization environment of BGFs. To maintain theoretical parsimony and consistency with the study objectives, the conceptual framework and empirical analysis focus specifically on the relationships among GEO, GMC, and SIEO.

Consistent with the study's objective of examining the roles of GEO and GMC in shaping SIEO, hypothesis development is restricted to these focal constructs. Other factors discussed in the literature, such as competitive intensity and operational capabilities, are considered part of the broader business environment and are not incorporated into the proposed model.

2.6 Hypotheses

This research examines how GEO and GMC jointly shape the SIEO of BGFs, drawing on the RBV and DCT. GEO represents a strategic behavioural capability that facilitates opportunity recognition and encourages innovation-driven internationalisation. In contrast, GMC reflects a flexible execution capability that enables firms to translate entrepreneurial intentions into sustained international success. Based on the conceptual framework illustrated in Figure 1, which links entrepreneurial orientation, managerial competence, and sustainability-oriented international outcomes, the subsequent hypotheses are projected:

H1: *GEO positively influences GMC.*
H2: *GMC has a positive effect on SIEO.*
H3: *GEO positively affects SIEO.*
H4: *GMC negatively moderates the relationship between GEO and SIEO.*

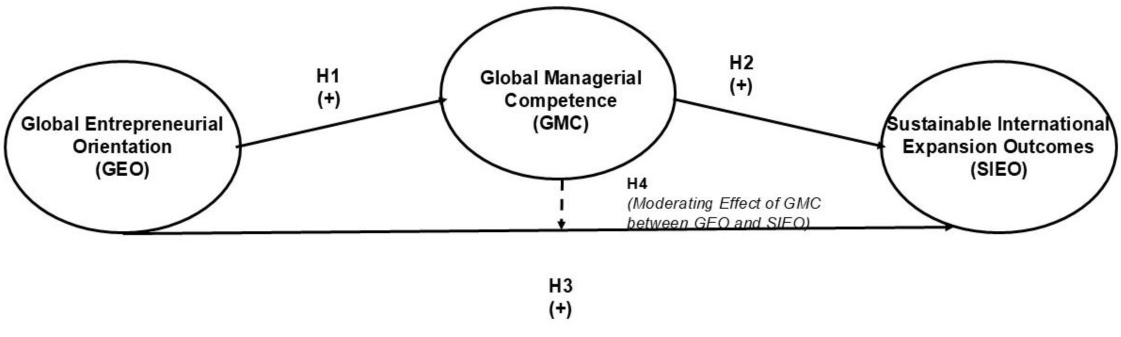

Figure 1. Conceptual framework integrating Global Entrepreneurial Orientation (GEO), Global Managerial Competence (GMC), and Sustainable International Expansion Outcomes (SIEO)

Figure 1 presents the integration of GEO, GMC, and SIEO within the frameworks of the RBV and DCT. According to RBV, entrepreneurial behaviour and managerial expertise represent critical intangible resources that support early internationalisation (Barney, 1991; Wernerfelt, 1984). However, RBV alone does not fully capture the adaptability required in fast-changing global markets. Dynamic Capability Theory, therefore, highlights the substance of evolving capacities that enable firms to sustain competitive advantage under conditions of uncertainty. In this context, born global enterprises require dynamic attributes, particularly GMC, to translate entrepreneurial intent into long-term international success.

3. Material and Methodology

3.1 Sample Procedure

This study develops and examines the proposed hypotheses regarding the relationship among GEO, GMC, and SIEO in the context of emerging market BGFs. Emerging market BGFs play a significant role in enhancing international trade, innovation, and economic development. Therefore, this study adopts a quantitative research approach using survey questionnaires and interviews to collect empirical data from BGFs operating in emerging economies.

The empirical investigation was conducted in three major stages. First, preliminary interviews were conducted with senior managers, founders, and export executives of selected BGFs to understand the practical dimensions of GEO and GMC. Second, a structured questionnaire was developed based on validated scales adopted from prior studies. Third, a large-scale survey was conducted through email distribution, online forms, and direct interaction with firm representatives.

The sampling framework was derived from export-oriented venture firms, startup databases, industrial associations, and government-supported export promotion organizations operating in emerging markets. The study targeted firms that were internationalized at an early stage and actively involved in foreign market operations.

A total of 450 questionnaires were distributed among BGFs. Out of these, 430 usable responses were received, resulting in an effective response rate of 95.56%. Consistent with previous studies on BGFs, the study operationalized BGFs as firms that were not older than five years and generated at least 40–45% of their total revenue from international markets (Jantunen et al., 2008; Knight & Cavusgil, 2004).

3.2 Measurement Variables and Reliability

The measurement scales used in this study were adapted from well-established prior research to ensure validity and reliability. All constructs were measured using a "5-point Likert Scale" ranging from "1 = strongly disagree" to "5 = strongly agree." To measure GEO, the study adapted the scale developed by Knight & Cavusgil (2004). GEO was measured through dimensions such as innovativeness, proactiveness, risk-taking behavior, and global market orientation. Nine measurement items were specifically designed to assess the extent to which firms actively pursue international entrepreneurial opportunities.

GMC was measured using scales adapted from prior international business and managerial competency studies (Martin & Javalgi, 2016; Peng & Luo, 2000). The construct evaluated managers' international business knowledge, strategic decision-making ability, cross-cultural communication skills, networking capability, and global market management competence. Respondents were asked to evaluate the effectiveness of managerial capabilities in handling international business operations and global expansion activities.

SIEO was measured using perceptual indicators adapted from Jantunen et al. (2008) and Hashai (2011). In this study, SIEO refers to the extent to which BGFs achieve sustainability-oriented outcomes through their international expansion activities. The construct captures environmentally sustainable practices in foreign markets, efficient resource utilisation across international operations, innovation-driven international expansion, reduction of environmental impacts associated with international business activities, and long-term stakeholder value creation. Respondents evaluated their firms' SIEO over the preceding three to five years.

To ensure the reliability and consistency of the constructs, Cronbach's Alpha, Composite Reliability (CR), and Average Variance Extracted (AVE) were examined. The results indicated satisfactory reliability and convergent validity values exceeding the recommended threshold levels suggested by Anderson & Gerbing (1988).

The collected data were analysed using Partial Least Squares Structural Equation Modelling (PLS-SEM) and moderation analysis to empirically test the proposed hypotheses and examine the moderating role of GMC in the relationship between GEO and SIEO.

4. Results and Discussion

4.1 Measures and Assessment of Reliability, Validity, and Common Method Bias

Table 1 presents the descriptive statistics for GEO, GMC, and SIEO. The mean scores indicate moderately high levels of GEO, GMC, and SIEO among the sampled BGFs. These descriptive results provide an initial overview of the focal constructs before assessment of the measurement and structural models.

Table 1. Descriptive statistics

Construct	Mean	SD
GEO	3.721	0.839
GMC	3.833	0.849
SIEO	3.967	0.829

Note: GEO: Global Entrepreneurial Orientation; GMC: Global Managerial Competence; SIEO: Sustainable International Expansion Outcomes; SD: Standard Deviation.

Table 1 presents the descriptive statistics for the main constructs used in this study, namely GEO, GMC, and SIEO. The mean score for GEO is 3.721 (Standard Deviation (SD) = 0.839), indicating that the sampled firms exhibit a moderately high level of entrepreneurial and international orientation. GMC has a mean of 3.833 (SD = 0.849), suggesting that managers in these firms possess strong global competencies. Finally, the mean for SIEO is 3.967 (SD = 0.829), indicating that firms demonstrate strong orientations toward achieving sustainable international growth.

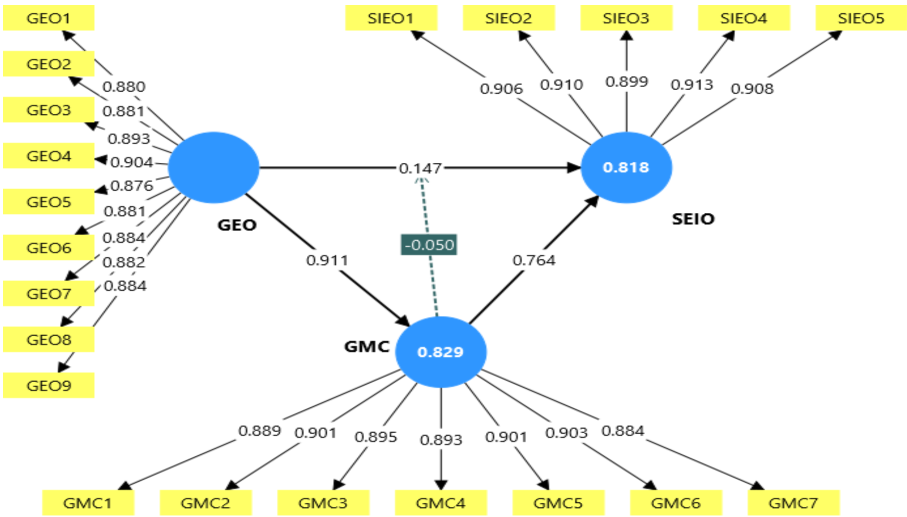

Figure 2. Structural model

Note: GEO: Global Entrepreneurial Orientation; GMC: Global Managerial Competence; SIEO: Sustainable International Expansion Outcomes.

Figure 2 presents the structural model examining the relationships among GEO, GMC, and SIEO. The results indicate that GEO has a strong positive effect on GMC with a path coefficient of $\beta = 0.911$, suggesting that firms with higher entrepreneurial orientation tend to develop stronger managerial competencies in international business operations. Furthermore, GMC demonstrates a substantial positive influence on SIEO ($\beta = 0.765$), implying that managerial competence significantly contributes to sustainable international entrepreneurial activities and long-term global expansion. The findings also reveal that GEO positively influences SIEO ($\beta = 0.147$), although the effect size is comparatively weaker than the effect of GMC. This indicates that entrepreneurial orientation directly contributes to sustainable international expansion while managerial competence remains a more dominant predictor. The moderating effect of GMC on the relationship between GEO and SIEO was found to be statistically significant but negative ($\beta = -0.051$, $p < 0.001$). This finding indicates that although GEO contributes positively to SIEO, its influence becomes weaker at higher levels of GMC.

A possible explanation is that highly competent managers tend to rely more heavily on formal planning, structured decision-making, and risk-management practices. Consequently, the additional benefits derived from entrepreneurial orientation become less pronounced when managerial competence is already strong. Thus, GMC functions as a buffering mechanism that reduces the dependence of international expansion outcomes on entrepreneurial orientation alone. The statistical significance of the interaction effect was further assessed through bootstrapping using the corresponding t-value, p-value, and 95% confidence interval. The model demonstrates acceptable explanatory power, with R^2 values of 0.829 for GMC and 0.818 for SIEO, indicating strong predictive capability of the proposed framework. According to Hair et al. (2021), R^2 values above 0.75 are considered substantial and indicate strong explanatory power. The model explains 82.9% of the variance in GMC and 81.8% of the variance in SIEO. According to the general interpretive guidelines for PLS-SEM, these R^2 values indicate substantial explanatory power for the endogenous constructs within the present sample (Hair et al., 2021). However, R^2 reflects the proportion of explained variance and should not, by itself, be interpreted as evidence of out-of-sample predictive performance or overall empirical validity. Predictive relevance is therefore assessed separately using Q^2 values and reported in Appendix B.

Table 2. Model fit

Saturated Model		Estimated Model
SRMR	0.022	0.022
dULS	0.112	0.114
dG	0.151	0.152
Chi-square	361.618	366.332
NFI	0.966	0.966

Note: NFI: Normed Fit Index.

Table 2 presents the model fit indices for the saturated and estimated models. The SRMR value of 0.022 is below the commonly applied threshold of 0.08, indicating a relatively small discrepancy between the observed and model-implied correlations (Hair et al., 2021). The dULS and dG values are also reported as discrepancy measures, while the NFI value of 0.966 indicates a comparatively high level of incremental fit. Taken together, these indices suggest an acceptable correspondence between the specified model and the observed data. Nevertheless, these fit statistics are interpreted as complementary diagnostic information and not as independent confirmation of the theoretical validity of the model.

Table 3. Correlation matrix of latent constructs

Construct	GEO	GMC	SIEO	GEO × GMC
GEO	1	-	-	-
GMC	0.911	1	-	-
SIEO	0.843	0.902	1	-
GEO × GMC	-0.004	-0.081	-0.108	1

Note: GEO: Global Entrepreneurial Orientation; GMC: Global Managerial Competence; SIEO: Sustainable International Expansion Outcomes; "-" denotes redundant entries for the upper triangle of symmetric correlation matrix and are not reported.

The correlation matrix (Table 3) indicates strong positive relationships among GEO, GMC, and SIEO, with the highest correlation observed between GMC and GEO ($r = 0.911$). However, the interaction term (GEO × GMC) shows very weak and negative correlations with the main constructs, suggesting a limited moderating interaction effect within the proposed model.

Table 4. Reliability and convergent validity

Construct	Cronbach's Alpha	Composite Reliability (CR)	AVE
GEO	0.965	0.970	0.783
GMC	0.959	0.966	0.801
SIEO	0.946	0.959	0.823

Note: GEO: Global Entrepreneurial Orientation; GMC: Global Managerial Competence; SIEO: Sustainable International Expansion Outcomes; AVE: Average Variance Extracted.

The reliability and validity results (Table 4) indicate that all constructs demonstrate excellent internal consistency and convergent validity. The Cronbach's Alpha and Composite Reliability values for GEO, GMC, and SIEO exceed the recommended threshold of 0.70, confirming strong construct reliability (Hair et al., 2021). Additionally, the AVE values are above 0.50, indicating adequate convergent validity and suggesting that the indicators explain a substantial proportion of variance in their respective constructs. Among the constructs, SIEO shows the highest AVE value (0.823), reflecting strong measurement quality and indicator representation.

Figure 3 represents the Heterotrait-Monotrait Ratio (HTMT) analysis, which is used to assess discriminant validity among the constructs in the proposed model. According to Henseler et al. (2015), HTMT values below 0.90 indicate acceptable discriminant validity, suggesting that the constructs are empirically distinct from one another. In the figure, the relationships among GMC–GEO, SIEO–GEO, and SIEO–GMC exhibit relatively high HTMT values, indicating strong conceptual associations among the constructs. However, these values remain within the acceptable threshold, confirming that each construct measures a distinct theoretical concept.

On the other hand, the interaction construct (GEO × GMC) demonstrates very low HTMT values with GEO, GMC, and SIEO. This indicates weak overlap and confirms that the moderating interaction term is statistically distinct from the primary constructs. The HTMT results confirm satisfactory discriminant validity and support the reliability of the measurement model.

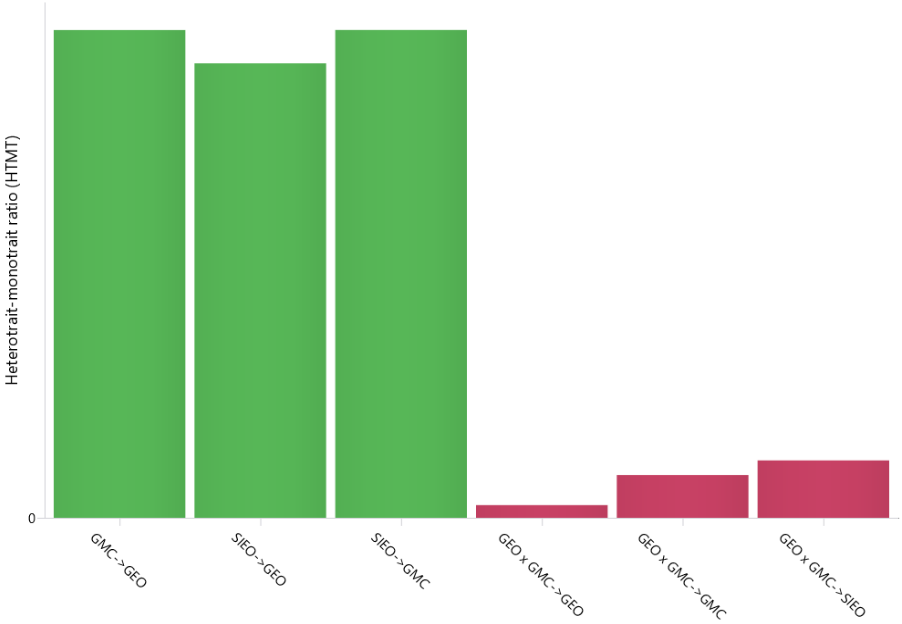

Figure 3. Heterotrait-Monotrait Ratio (HTMT) values for discriminant validity assessment

Table 5. Hypothesis testing results

Hypothesis	Path	β	*t*-Value	*p*-Value	Result
H1	GEO → GMC	0.911	133.888	<0.001	Supported
H2	GMC → SIEO	0.765	16.205	<0.001	Supported
H3	GEO → SIEO	0.147	2.964	<0.001	Supported
H4	GMC x GEO → SIEO	-0.051	2.449	<0.001	Supported

Note: GEO: Global Entrepreneurial Orientation; GMC: Global Managerial Competence; SIEO: Sustainable International Expansion Outcomes.

Table 5 presents the results of hypothesis testing for the proposed structural model. The findings reveal that GEO has a significant positive effect on GMC (β = 0.911, t = 133.888, p < 0.001). The exceptionally high beta coefficient indicates that firms possessing strong entrepreneurial orientation at the global level are more likely to develop superior managerial competencies required for international business operations. Therefore, H1 is supported.

In addition, the relationship between GMC and SIEO was found to be positive and statistically significant (β = 0.765, t = 16.205, p < 0.001). The result demonstrates that managerial competencies play a critical role in strengthening sustainable international expansion and long-term global business performance. Managers possessing international knowledge, strategic decision-making capabilities, and cross-cultural management skills contribute substantially to the sustainable growth of BGFs. Therefore, H2 is supported.

The results further indicate that GEO positively influences SIEO (β = 0.147, t = 2.964, p < 0.001). Although the magnitude of the relationship is relatively moderate, the association remains statistically significant. This finding suggests that firms with higher GEO tend to achieve better SIEO. Hence, H3 is supported.

Finally, the interaction effect of GMC and GEO on SIEO was found to be statistically significant (β = −0.051, t = 2.449, p < 0.001). Although the interaction coefficient is negative, its magnitude is relatively small, indicating a weak moderating effect. The result suggests that the combined influence of GEO and GMC may create certain managerial or strategic complexities during international expansion. Nevertheless, the relationship remains statistically significant, leading to the acceptance of H4. The positive effect of GEO on SIEO becomes weaker as GMC increases. Therefore, GMC acts as a buffering factor rather than an enhancing factor in the relationship between GEO and SIEO.

Figure 4 illustrates the moderating role of GMC in the relationship between GEO and SIEO. The positive slopes of both lines indicate that GEO contributes positively to SIEO under both low and high levels of GMC. However, the slope associated with high GMC is noticeably flatter than that of low GMC. This pattern suggests a negative moderation effect, consistent with the significant negative interaction coefficient (β = −0.051, p < 0.001). Specifically, although firms with stronger entrepreneurial orientation tend to achieve better international expansion outcomes, the magnitude of this positive relationship decreases as managerial competence increases. A plausible explanation is that highly competent managers rely more heavily on structured planning, formal decision-making processes, and risk-management capabilities. As a result, the incremental contribution of entrepreneurial orientation becomes less critical for achieving SIEO. Conversely, when managerial competence is relatively low, entrepreneurial orientation plays a more prominent role in driving international expansion success. Therefore, the interaction plot confirms that GMC does not strengthen the GEO–SIEO relationship as originally hypothesized. Instead, GMC acts as a buffering factor that weakens the positive influence of GEO on SIEO. Accordingly, H4 receives partial support because the moderation effect is statistically significant but opposite to the hypothesized positive direction.

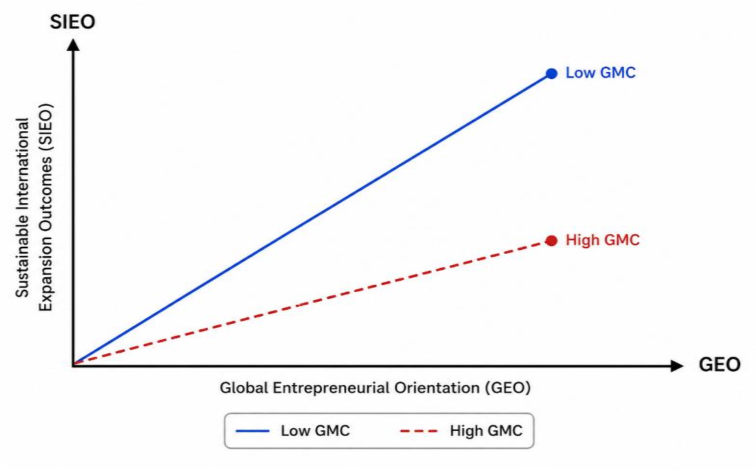

Figure 4. Negative moderation of Global Managerial Competence (GMC) in the Global Entrepreneurial Orientation (GEO)–Sustainable International Expansion Outcomes (SIEO) relationship

Table 6. Multicollinearity test VIF

Hypotheses	VIF
GEO → SIEO	2.13
GMC → GEO	1.00
GEO × GMC → SIEO	1.04

Note: GEO: Global Entrepreneurial Orientation; GMC: Global Managerial Competence; SIEO: Sustainable International Expansion Outcomes; VIF: variance inflation factor.

As shown in Table 6, the variance inflation factor (VIF) values were assessed to examine the issue of multicollinearity among the predictor constructs in the structural model. According to Hair et al. (2021), in the present model, the VIF value for the GEO → SIEO path is 2.13, which is well below the recommended threshold of 5. Therefore, no serious multicollinearity concern was detected in the structural model. The remaining VIF values are also below the recommended threshold, further confirming the absence of multicollinearity issues. However, the interaction term GEO × GMC → SIEO shows a low VIF value of 1.04, indicating no multicollinearity problem for the moderating effect. Similarly, the GMC→ GEO path has a VIF value of 1.00, reflecting the complete absence of collinearity issues.

4.2 Structural Model Assessment and Moderation Analysis

Following assessment of the measurement model, the structural model was evaluated using path coefficients, *t*-values, *p*-values, coefficients of determination (R^2), VIF, predictive relevance (Q^2), effect sizes (f^2), and bootstrapped 95% confidence intervals. The reliability and convergent validity results reported in Table 4 indicate satisfactory measurement quality, while the HTMT assessment presented in Figure 3 provides evidence regarding discriminant validity. Collinearity was assessed using VIF values, as reported in Table 6. The structural relationships were subsequently examined through bootstrapping, with the path coefficients, *t*-values, and *p*-values presented in Table 5. Additional structural model statistics, including Q^2, f^2, and bootstrapped confidence intervals, are reported in Appendix B, Tables B1–B3. These results provide a solid foundation for interpreting the structural model. The explanatory and predictive assessments provide additional information regarding the structural model. The R^2 values of 0.829 for GMC and 0.818 for SIEO indicate that the model explains a substantial proportion of the variance in the two endogenous constructs. Predictive relevance was assessed using Q^2, with values of 0.641 for GMC and 0.587 for SIEO; both values are above zero, indicating predictive relevance for the endogenous constructs. The f^2 results further show that GEO has a large effect on GMC ($f^2 = 4.950$), while GMC has a large effect on SIEO ($f^2 = 0.742$). In comparison, the direct effect of GEO on SIEO is small ($f^2 = 0.028$), and the GEO × GMC interaction contributes a small incremental effect ($f^2 = 0.012$). Finally, the bootstrapped 95% confidence intervals for GEO → GMC [0.897, 0.924], GMC → SIEO [0.673, 0.842], GEO → SIEO [0.052, 0.241], and GEO × GMC → SIEO [−0.092, −0.011] do not include zero. Collectively, these results support the statistical significance of the estimated structural relationships while also indicating that the moderation effect is small in magnitude. Detailed results are provided in Appendix B (Tables B1–B3). The hypothesis-testing results reveal several important relationships among the constructs.

Table 5 presents the results of hypothesis testing for the proposed structural relationships among GEO, GMC, and SIEO. The findings reveal that GEO has a strong and significant positive effect on GMC ($\beta = 0.911$, $t = 133.888$, $p < 0.001$). This result supports H1 and indicates that firms with a higher level of entrepreneurial orientation are more likely to develop stronger managerial competencies in international markets.

The results further demonstrate that GEO positively influences SIEO ($\beta = 0.147$, $t = 2.964$, $p < 0.001$). Although the effect size is comparatively small, the relationship remains statistically significant, thereby supporting H3. This suggests that entrepreneurial orientation contributes positively to sustainable international entrepreneurial activities and global market expansion.

In addition, GMC shows a strong positive impact on SIEO ($\beta = 0.765$, $t = 16.205$, $p < 0.001$), providing support for H2. The findings indicate that managerial competence plays a critical role in enhancing SIEO and long-term international growth.

Finally, the interaction effect between GMC and GEO on SIEO is significant ($\beta = -0.051$, $t = 2.449$, $p < 0.001$), supporting H4. However, the negative coefficient suggests that the moderating or interaction effect slightly weakens the direct relationship with SIEO. Despite this negative sign, the relationship remains statistically significant, indicating that the combined influence of managerial competence and entrepreneurial orientation affects sustainable international entrepreneurial outcomes. Appendix A presents Questionnaire Items and Appendix B, including Q^2 statistics, f^2 effect sizes, and bootstrapped confidence intervals.

5. Conclusions

This study examined the influence of GEO and GMC on SIEO among BGFs operating in emerging economies. Drawing upon the RBV and DCT, the study provides a comprehensive understanding of how entrepreneurial orientation and managerial capabilities collectively contribute to sustainable global expansion and long-term international competitiveness. The findings demonstrate that GEO exerts a strong and significant positive impact on GMC, indicating that firms with higher levels of innovativeness, proactiveness, and risk-taking behavior are more likely to develop globally competent managerial capabilities. Entrepreneurially oriented firms actively identify international opportunities, adapt to rapidly changing global environments, and pursue innovation-driven strategies that enhance international market performance. The results further confirm that GEO positively influences SIEO, suggesting that firms with strong GEO are better positioned to achieve sustainable international growth and long-term market sustainability. In addition, GMC was found to have a substantial positive effect on SIEO, highlighting the critical importance of managerial capabilities in global business success. Managers possessing international business expertise, strategic decision-making abilities, networking competence, and cross-cultural management skills significantly contribute to sustainable international expansion and organizational resilience. The moderation analysis further showed that GMC significantly conditions the relationship between GEO and SIEO. Specifically, the negative interaction coefficient indicates that the positive influence of GEO on SIEO becomes weaker as GMC increases. This finding suggests a buffering rather than an enhancing moderation effect. In firms with comparatively lower managerial competence, entrepreneurial orientation appears to play a more prominent role in supporting sustainable international expansion, whereas highly competent managers may rely more extensively on structured planning, international experience, and formal decision-making capabilities. Consequently, the incremental contribution of GEO becomes less pronounced at higher levels of GMC. Overall, the findings provide empirical support for the proposed relationships among GEO, GMC, and SIEO within the present sample of emerging-market BGFs. The model explains a substantial proportion of variance in the endogenous constructs and demonstrates predictive relevance based on the reported Q^2 values. However, the relatively small effect size of the interaction term indicates that the moderating influence of GMC should be interpreted cautiously. The study therefore contributes to the international entrepreneurship literature by offering evidence on the joint role of entrepreneurial orientation and managerial competence in sustainable international expansion, while recognising the contextual and cross-sectional boundaries of the reported findings.

5.1 Limitations and Future Research Directions

Despite its significant contributions, this study has certain limitations that provide opportunities for future research. First, the study was limited to BGFs operating in selected emerging economies, namely India, Malaysia, Singapore, and the United Arab Emirates. Therefore, the findings may not be fully generalizable to firms operating in developed economies or other regional contexts with different institutional and economic environments. Second, the study adopted a cross-sectional research design, which captured data at a single point in time. As international entrepreneurial behavior and managerial competencies evolve, future studies may adopt longitudinal research designs to examine the dynamic changes in GEO, GMC, and SIEO over different stages of firm growth and internationalization. Third, the study primarily relied on self-reported survey responses collected from managers and firm representatives, which may increase the possibility of common method bias and subjective interpretation. Future research may incorporate secondary financial data, case studies, or mixed-method approaches to improve the robustness and validity of findings.

Furthermore, the present study focused on the moderating role of GMC in the GEO–SIEO relationship. Future research may examine additional boundary conditions, such as innovation capability, digital transformation, organisational resilience, institutional support, environmental uncertainty, and competitive intensity, to determine whether the strength and direction of the GEO–SIEO relationship vary across different organisational and environmental contexts. Future studies may examine additional mediating or moderating variables such as innovation capability, digital transformation, organizational resilience, institutional support, environmental uncertainty, and competitive intensity to develop a more comprehensive understanding of sustainable global expansion. Finally, future research may extend the proposed framework by integrating sustainability dimensions such as green innovation, ESG practices, circular economy strategies, and digital internationalization in alignment with SDGs. Such investigations would further enrich the literature on international entrepreneurship and sustainable global business development.

5.2 Policy Implication

The findings of this study provide several important policy implications for governments, policymakers, industry associations, and entrepreneurial support institutions in emerging economies. Since GEO and GMC significantly contribute to SIEO, policymakers should prioritize the development of globally competitive entrepreneurial ecosystems that encourage innovation, internationalization, and sustainable business growth. First, governments should design and implement policies that support innovation-driven entrepreneurship and facilitate the early internationalization of BGFs. Financial incentives, export promotion programs, startup funding schemes, tax benefits, and simplified international trade regulations can help emerging firms expand into foreign markets more effectively. Such initiatives can strengthen firms' entrepreneurial orientation and improve their global competitiveness. Second, the study highlights the strategic importance of managerial competence in achieving sustainable international expansion. Therefore, policymakers and educational institutions should promote international business education, managerial training programs, leadership development initiatives, and cross-cultural competency enhancement programs. Building globally competent managers will enable firms to manage international uncertainty, global competition, and cross-border business operations more efficiently. Third, governments and industrial organizations should strengthen international networking platforms, digital infrastructure, and innovation support systems to facilitate collaboration between firms, foreign investors, research institutions, and international business partners. Such support can improve firms' access to international knowledge, technology transfer, and global market opportunities. Furthermore, policymakers should integrate sustainability-oriented policies aligned with SDG-9, focusing on innovation, resilient infrastructure, and sustainable industrialization. Encouraging green innovation, digital transformation, and environmentally responsible business practices can help BGFs achieve long-term sustainable global expansion.

Author Contributions

Conceptualization, N.K and F.R.A.; methodology, A.H.; software, S.A.; validation, A.I.; formal analysis, N.K.; investigation, A.H.; resources, S.A.; data curation, A.I.; writing—original draft preparation, N.K.; writing—review and editing, S.K.C.; visualization, N.K.; supervision, S.K.C.; project administration, S.K.C.; funding acquisition, N.K. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Ethical Approval

This study was conducted in accordance with standard ethical research practices for survey-based academic research involving voluntary participation. The data were collected from founders, CEOs, and managers of international firms through structured questionnaires. Prior to participation, all respondents were informed about the academic purpose of the study, and informed consent was obtained from each participant.

The participation was entirely voluntary, and respondents were assured that their identities and responses would remain anonymous and confidential. No personally identifiable information was collected or disclosed during the research process. According to the institutional guidelines applicable to the authors' affiliated institution, formal ethical approval was not required for non-clinical and anonymous survey-based research involving adult professional participants.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors utilized Quillbot for language correction. Afterward, they reviewed and edited the content as necessary and took full responsibility for the publication's content.

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Appendix

Appendix A.

Research Questionnaire

All items were assessed using a five-point Likert scale, where 1 indicates Strongly Disagree and 5 indicates Strongly Agree.

Table A1. Global Entrepreneurial Orientation (GEO)

Code	Measurement Items
GEO1	Our firm places strong emphasis on innovation in products and services for international markets.
GEO2	We proactively identify and pursue new international market opportunities before competitors.
GEO3	Our firm demonstrates a willingness to undertake significant risks in foreign markets.
GEO4	We regularly adopt and implement new technologies to enhance global competitiveness.
GEO5	Our international strategies exhibit a high level of proactiveness.
GEO6	Our firm is willing to commit significant resources to international ventures even when outcomes are uncertain.
GEO7	Innovation plays a central role in driving our growth in international markets.
GEO8	Our firm actively experiments with new business models when entering international markets.
GEO9	We regularly try out new ideas and approaches when operating in international markets.

Table A2. Global Managerial Competence (GMC)

Code	Measurement Items
GMC1	Managers in this firm communicate effectively across different cultural contexts.
GMC2	Managers in this firm have substantial hands-on experience in international business activities.
GMC3	Managers in this firm are able to adapt strategies effectively to suit foreign market conditions.
GMC4	Managers in this firm maintain strong and well-developed professional networks at the global level.
GMC5	Managers in this firm make sound strategic decisions in international environments.
GMC6	Managers in this firm are capable of handling uncertainty effectively in international operations.
GMC7	Managers in this firm learn promptly from their experiences in global markets.

Table A3. Sustainable International Expansion Outcomes (SIEO)

Code	Measurement Items
SIEO1	Efficient utilization of resources in foreign expansion
SIEO2	Socially responsible engagement in international markets, and long-term sustainable value creation
SIEO3	Environmentally sustainable international operations
SIEO4	Contribution to sustainable industrial development
SIEO5	Adoption of green and innovative business practices

Appendix B.

Table B1 shows that Q^2 values are greater than zero, indicating satisfactory predictive relevance of the structural model.

Table B1. Predictive Relevance (Q^2)

Construct	Q^2
GMC	0.641
SIEO	0.587

Note: GMC = Global Managerial Competence; SIEO = Sustainable International Expansion Outcomes.

Table B2, GMC exerts a substantial effect on SIEO, whereas the moderation effect contributes only a small incremental effect.

Table B2. Effect Size (f^2)

Relationship	f^2	Effect Size
GEO → GMC	4.95	Large
GEO → SIEO	0.028	Small
GMC → SIEO	0.742	Large
GEO × GMC → SIEO	0.012	Small

Note: GEO = Global Entrepreneurial Orientation; GMC = Global Managerial Competence; SIEO = Sustainable International Expansion Outcomes.

Table B3: None of the confidence intervals include zero, confirming the statistical significance of the hypothesized relationships.

Table B3. Bootstrapped confidence intervals (95%)

Path	β	LLCI	ULCI
GEO → GMC	0.911	0.897	0.924
GMC → SIEO	0.765	0.673	0.842
GEO → SIEO	0.147	0.052	0.241
GEO × GMC → SIEO	-0.051	-0.092	-0.011

Note: LLCI = lower limit of 95% bias-corrected bootstrap confidence interval; ULCI = upper limit of 95% bias-corrected bootstrap confidence interval; GEO = Global Entrepreneurial Orientation; GMC = Global Managerial Competence; SIEO = Sustainable International Expansion Outcomes.

The predictive relevance assessment revealed Q^2 values of 0.641 for GMC and 0.587 for SIEO, indicating satisfactory predictive capability of the model. The effect size analysis showed that GEO had a substantial impact on GMC, while GMC exerted a large effect on SIEO. In contrast, the interaction term (GEO × GMC) exhibited a relatively small effect size, suggesting a weak but statistically significant moderating influence. Furthermore, bootstrapped 95% confidence intervals did not include zero for any of the structural paths, thereby confirming the robustness and significance of the estimated relationships.